F1 PERFORMANCE ANALYSIS REPORT

Exploratory Data Analysis of Driver, Constructor and Race Performance

Dataset: f1_performance_analysis_sample_720(1).csv
Coverage: 2022–2024 | **Records:** 720 | **Drivers:** 20 | **Constructors:** 10 | **Race events:** 36

1. Project Summary

This report analyzes 720 F1 performance records across 36 race events from 2022–2024. The dataset contains 4,135 recorded points. Max Verstappen has the highest driver total (594), and Red Bull has the highest constructor total (1,097). The analysis covers driver performance, constructor performance, season trends, qualifying versus finishing, position gains, DNFs and data quality.

Data-quality checks found 111 missing cells, all 37 in Positions_Gained, and 0 duplicate rows. The Is_Win field records multiple winners in 24 races, while Is_Podium exceeds three in 15 races. These fields should therefore be validated before being used as formal KPIs.

2. Dataset and Cleaning

Metric	Value
Records	**720**
Columns	**23**
Drivers	**20**
Constructors	**10**
Race events	**36**
Total points	**4,135**
Missing cells	**111**
Duplicate rows	**0**

Finish_Position was converted to a numeric analysis field, and Position_Gained_Calc was derived as Grid_Position minus numeric Finish_Position. DNF/DNS entries do not receive a numeric finishing position.

3. Key Analysis Results

Top driver: Max Verstappen — 594 points.

Top constructor: Red Bull — 1,097 points.

Highest-point race: Miami GP (2022) — 166 points.

Qualifying/finish correlation: 0.81. This indicates a strong relationship in the sample, not causation.

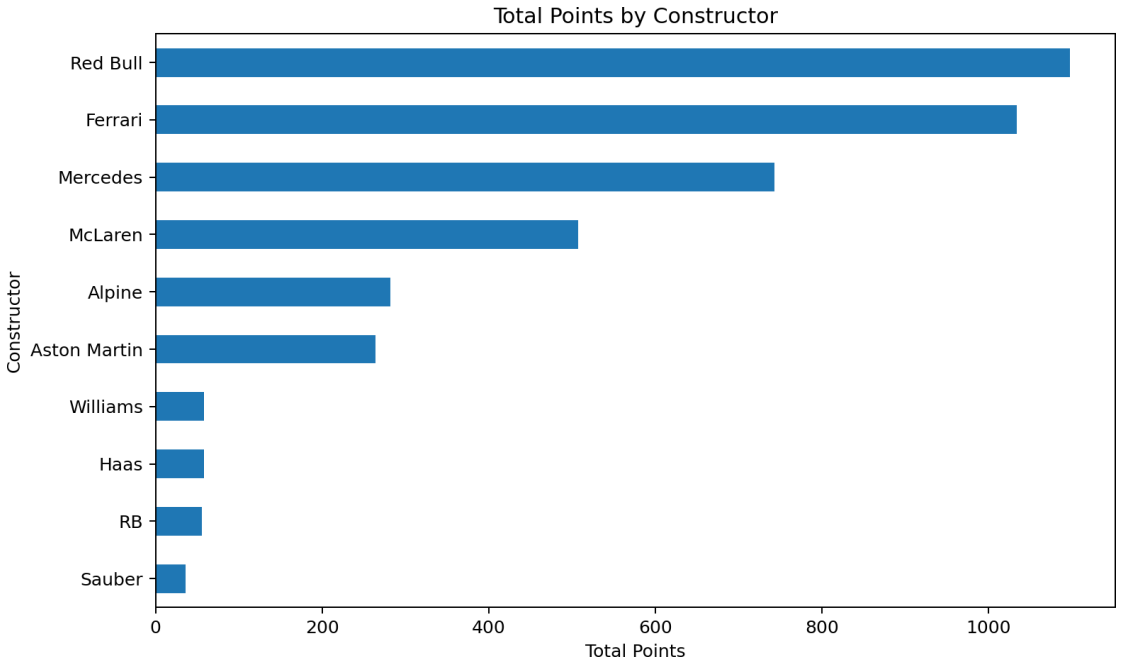
Highest average position gain: Esteban Ocon — 1.49 positions.

4. Season Results

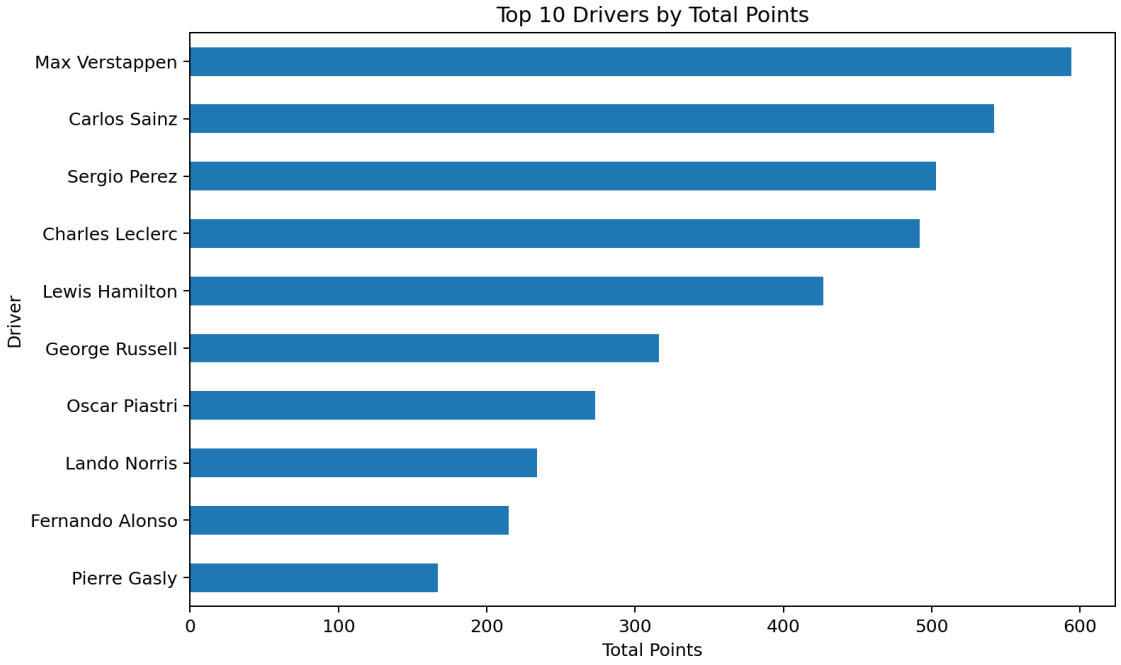
Season	Points	Wins*	Podiums*	DNFs
2022	1525	29	49	**16**
2023	1264	24	40	**13**
2024	**1346**	**24**	**40**	**8**

*Wins and podium totals are shown as recorded but are subject to the data-quality limitation described above.

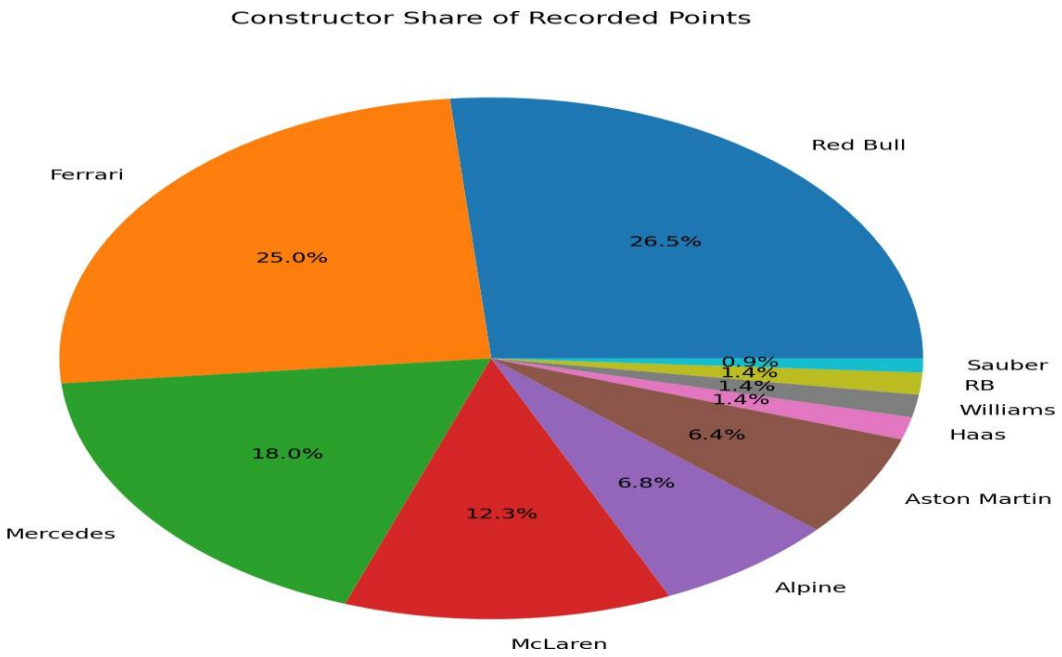
5. Visual Analysis



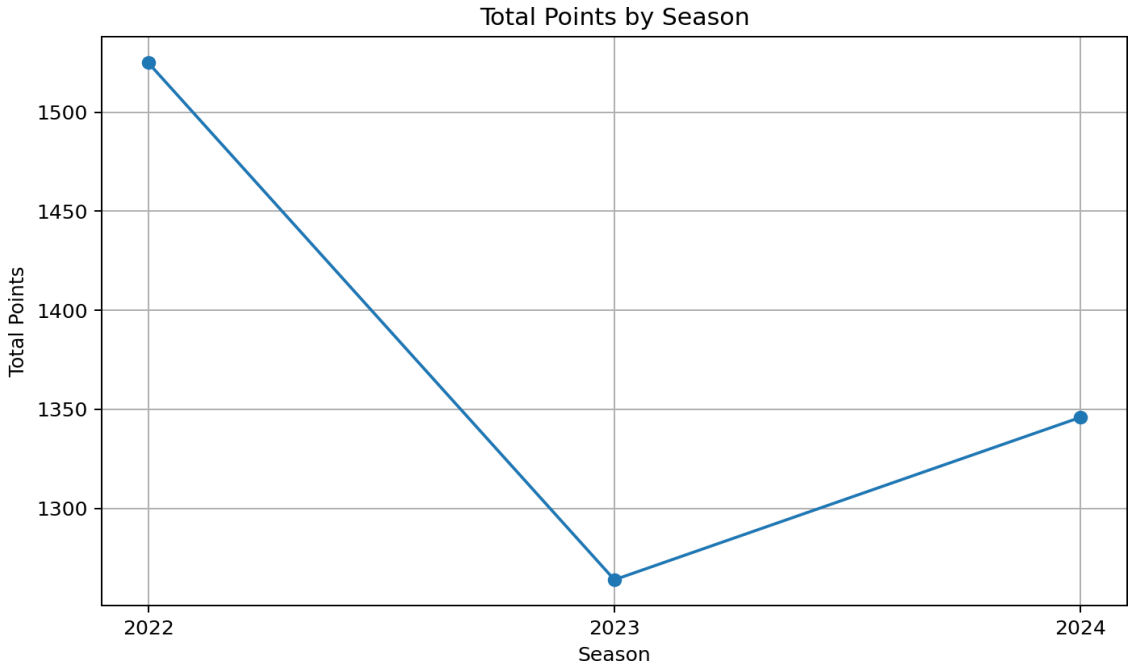
Total Points by Constructor



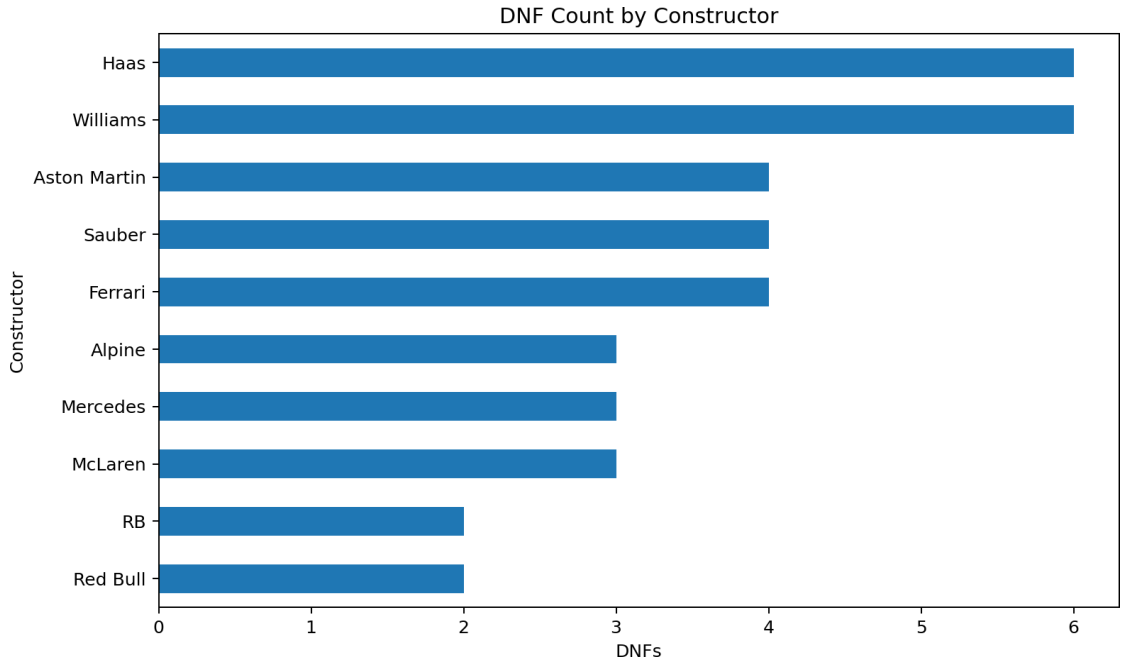
Top 10 Drivers by Total Points



Constructor Share of Recorded Points



Total Points by Season



DNF Count by Constructor

6. Key Insights

• Max Verstappen has the highest recorded driver points total at 594.

• Red Bull has the highest recorded constructor points total at 1,097.

• The dataset records 4,135 total points across 36 race events.

• Recorded DNFs fall from 16 in 2022 to 8 in 2024, indicating fewer non-finishing outcomes in the later season within this sample.

• Qualifying and finishing positions have a correlation of 0.81, showing a strong relationship in the sample between starting performance and race finishing performance.

• Esteban Ocon has the highest average calculated position gain, at approximately 1.49 positions per race.

• The dataset's recorded points peak at the Miami GP (2022), with 166 points.

• The binary Is_Win and Is_Podium fields should not be used as sole performance measures because their race-level totals are inconsistent with normal F1 rules.

7. Recommendations

• Prioritize qualifying performance analysis because qualifying position is strongly associated with finishing position in this dataset.

• Investigate races and constructors with higher DNF counts to identify reliability or operational patterns that may reduce completed-race performance.

• Use position-gain analysis to study successful recovery strategies, especially for drivers who frequently finish ahead of their starting grid position.

• For future dashboards, prioritize Points, Finish_Position, Grid_Position and Status as core performance measures and use Is_Win/Is_Podium only after validation.

• Clean and validate the binary indicators at source so that each race has one winner and no more than three podium finishers before using those fields for KPI reporting.

8. Limitations and Conclusion

The dataset is best treated as a supplied analytical sample rather than an authoritative historical F1 database. The race calendar is partial, Positions_Gained has missing values, and binary win/podium indicators are internally inconsistent. Despite these limitations, the points, finishing-position, qualifying and DNF fields provide useful descriptive performance analysis. The recommended next step is to validate the event-level outcome flags and then build a repeatable F1 performance dashboard.